ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** -1 D	P = 1/(-1 m).
2	**(b)** a single red patch	Monochromatic light does not disperse.
3	**(b)** dark	No scattering.
4	**(b)** +2 D	1/f = 1/25 - 1/50 = 1/50 gives +2 D.
5	**(d)** the refractive index for violet is higher	n is larger for shorter wavelengths.
6	**(b)** light travels a short path and little blue scatters	Colours combine.
7	**(a)** -40 cm	f = 1/P.
8	**(a)** red	n is smallest for red.
9	**(a)** blue light	Short wavelength scatters more.
10	**(c)** in front of the retina	Because the eye is too strong.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	1/f = -1/40 + 1/25 = 3/200 (1); f = 66.7 cm (1).	1+1
12	His near point is beyond 25 cm (1); Only at the larger distance the image falls on the retina (1).	1+1
13	Less air above scatters less light (1); The sky looks deep blue/dark (1).	1+1
14	f = -2 m (1); P = -0.5 D (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	u = infinity, v = -1.5 m (1); f = -1.5 m (1); P = -0.67 D, concave lens (1).	1+1+1
16	At noon light has a short path in the atmosphere (1); At sunset it travels a long path (1); Most blue is scattered away, so red remains (1).	1+1+1
17	25 cm (1); At smaller distances the lens cannot become thick enough (1); The image would be blurred (1).	1+1+1
	OR: 1/f = -1/75 + 1/25 = 2/75 (1); f = 37.5 cm (1); P = +2.67 D (1).	1+1+1
18	The second prism reverses the dispersion (1); Colours combine to form white light (1); Shows the prism only separates existing colours (1).	1+1+1
	OR: They are near, so they appear as extended sources (1); Variation of light from a point on them is averaged out (1); Atmospheric refraction effects cancel (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(a) Hypermetropia (1); (b) 1/f = -1/100 + 1/25 = 3/100 (1); f = 33.3 cm, P = +3 D (1); (c) Convex lens (1); Image forms at the near point (1).	1+1+1+1+1
	OR: u = infinity, v = -80 cm (1); f = -80 cm (1); P = -1.25 D (1); Near point is 25 cm, normal (1); No lens needed for reading, but a concave lens for distance (1).	1+1+1+1+1
20	(a) Myopia (1); (b) u = infinity, v = -40 cm (1); f = -40 cm (1); P = -2.5 D (1); (c) Concave lens diverges rays so a distant object appears at 40 cm (1).	1+1+1+1+1
	OR: Dispersion: spectrum from a prism (1); Scattering: blue sky (1); Refraction: pencil bent in water (1); Atmospheric refraction: twinkling of stars (1); Examples are drawn from daily life (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.

- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.
- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** white light	Recombination.
2	**(c)** of atmospheric refraction	Rays are bent differently.
3	**(c)** the focal length of the lens	Accommodation.
4	**(a)** the Sun is behind the observer and water drops are in front	Sunlight must fall on raindrops in front of the observer.
5	**(b)** a clear night through a turbulent atmosphere	Because air density changes.
6	**(c)** relax	The lens becomes thinner.
7	**(c)** red	Violet is lowest.
8	**(c)** darker blue	Less scattering.
9	**(a)** the pupil takes time to dilate	Iris adjusts slowly.
10	**(d)** diffraction by the sun	Diffraction is not involved.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The two faces are parallel (1); The second face undoes the dispersion (1).	1+1
12	Red scatters the least (1); It is seen from a long distance (1).	1+1
13	Upper part is concave for distant vision (1); Lower part is convex for near vision (1).	1+1
14	Light from the droplets is returned back towards the Sun (1); The observer must stand between the Sun and the rain (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	f = -25 cm (1); Far point is at 25 cm (1); Defect is myopia (1).	1+1+1
16	Dust and smoke particles in the room scatter the light (1); This is the Tyndall effect (1); Scattered light reaches the observer's eye (1).	1+1+1
17	Light refracts on entering a droplet (1); It disperses into colours and reflects internally (1); It refracts again on leaving (1).	1+1+1
	OR: f = 40 cm (1); 1/v = 1/f + 1/u = 1/40 - 1/25 = -3/200 (1); v = -66.7 cm, so near point is 66.7 cm (1).	1+1+1
18	Scattered sunlight lights the sky and drowns the star (1); The star's light is much weaker (1); The eye cannot distinguish it (1).	1+1+1
	OR: f = -50 cm (1); Far point at 50 cm (1); Image formed at 50 cm by the lens (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lens system is the cornea and eye lens (1); Iris and pupil act as the aperture (1); Retina acts as the film (1); Ciliary muscles focus the image (1); Optic nerve transmits the image signals to the brain (1).	1+1+1+1+1
	OR: Layers of air have different refractive indices (1); Starlight bends towards the normal as it enters denser air (1); Apparent position and brightness vary (1); Planets are extended sources (1); Sun seen about 2 minutes before sunrise and after sunset (1).	1+1+1+1+1
20	A narrow beam is passed through a prism (1); Spectrum of seven colours is obtained on the screen (1); A second inverted prism is placed (1); White light is recovered (1); A single colour does not break up further, so the prism does not add colour (1).	1+1+1+1+1
	OR: (a) Presbyopia with myopia (1); (b) f = 50 cm for +2 D (1); f = -66.7 cm for -1.5 D (1); (c) Bifocal lens (1); Upper part concave, lower part convex (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that a myopic eye is corrected with a convex lens, or hypermetropia with a concave lens; myopia needs concave and hypermetropia needs convex.
- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
- Writing that the image on the retina is erect; it is real and inverted and the brain interprets it as erect.
- Stating that red light is scattered the most; it is scattered the least, and blue light is scattered the most.
- Forgetting the negative sign for a myopia lens, or using f in centimetres to find power; $P = 1/f$ with f in metres.
- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** Advance sunrise	Sun seen before it rises.
2	**(d)** from 25 cm to infinity	Range of distinct vision.
3	**(a)** a beam of pure red light	Monochromatic light does not disperse.
4	**(c)** smoke particles scatter light	Tyndall effect.
5	**(d)** Presbyopia	Weakening of muscles.
6	**(c)** Red	Red.
7	**(a)** it is scattered less	Longer wavelengths penetrate better.
8	**(a)** convex lenses of power +3 D	1/f = -1/100 + 1/25.
9	**(a)** no dispersion	Emerging faces are parallel.
10	**(c)** Violet and blue	Shortest wavelengths.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Convex lens would converge the rays more (1); The image would form farther in front of the retina (1).	1+1
12	Ciliary muscles contract and increase the curvature (1); A more curved lens has a shorter focal length (1).	1+1
13	Light must travel through thicker air (1); Most blue is scattered out (1).	1+1
14	Atmospheric refraction near the horizon (1); Rays from the lower edge bend more than the upper (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use bifocal lenses (1); Upper part concave, lower part convex (1); Convenient for both distant and near vision (1).	1+1+1
16	In bright light the iris constricts the pupil (1); Less light enters (1); Retina is protected (1).	1+1+1
17	Presbyopia and cataract in old age (1); Presbyopia: bifocal lenses (1); Cataract: replacement of the lens by surgery (1).	1+1+1
	OR: Dispersion: splitting into colours by refraction (1); Scattering: deflection by particles (1); Examples: rainbow and blue sky (1).	1+1+1
18	Image forms in front of the retina (1); Myopia (1); Concave lens (1).	1+1+1
	OR: Blue is scattered the most (1); It would not be seen from a long distance (1); Red passes through fog better (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	The observer faces west (the Sun is behind) (1); Raindrops are in the west (1); Light refracts, disperses and reflects inside drops (1); Light leaves the drops towards the observer (1); Red on top, violet at the bottom (1).	1+1+1+1+1
	OR: Myopia (1); Eyeball too long or lens too curved (1); f = -6 m (1); P = -0.167 D (1); Concave lens (1).	1+1+1+1+1
20	(a) Small particles scatter blue the most (1); (b) Large droplets scatter all colours equally (1); (b) Equal scattering gives white (1); (c) Long path through the atmosphere (1); (c) Red is least scattered and reaches the eye (1).	1+1+1+1+1
	OR: Tyndall effect: scattering by colloidal particles (1); Dispersion: splitting into colours by a prism (1); Tyndall effect makes the beam visible (1); Dispersion forms a spectrum (1); Examples: smoke room beam and rainbow (1).	1+1+1+1+1

Mistakes to avoid:

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- Believing that most refraction in the eye occurs at the eye lens; it occurs at the outer surface of the cornea, and the lens only fine-tunes the focal length.
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- Saying that planets twinkle like stars; stars twinkle due to atmospheric refraction of a point source, whereas planets appear as extended sources.

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